Feature extraction methods

In the latest study, we annotated 8,791 posts from 1,178 users. Each user was tasked with annotating 20 unique posts.

Method 1: BERTopic

BERTopic is a topic modeling method that treats topic discovery as a pipeline. That pipeline has the following steps:

1. Embed documents: Convert documents to high-dimensional embeddings using transformer models like BERT.
2. Reduce dimensionality: Apply UMAP to reduce embeddings to 2D or 3D for effective clustering.
3. Cluster the reduced embeddings: Use HDBSCAN or similar algorithms to group similar documents.
4. Extract topic representations: Generate topic labels by identifying the most important words in each cluster.

BERTopic collapses the task of feature discovery into two tasks:

1. Generate groups of similar items (via clustering).
2. Generate descriptions of each group (via topic representation).

We leveraged BERTopic in order to generate features from the original and mirrored post stimuli. We generated embeddings for all posts (n=17,582), clustered them (using both HDBSCAN and K-Means), and generated human-readable labels using AI.

Using this approach, we derived n=53 topics across the dataset. We filter features that are artifacts of the keywords that were used during ingestion. We also did other filtering related to duplication and quality. We then arrived at a list of top n=20 features.

Number of posts	LLM-generated label
210	Criticism of MAGA and related political beliefs
201	Criticism of Democrats vs Republicans (DNC/MAGA) and claims of dishonesty and weak agendas
200	Criticism of Republicans and GOP policies/actions
179	Deportation and illegal immigration debate
145	Biden and Democratic immigration policy criticism
143	Political debate and criticism of left vs. right wing rhetoric
129	Trump corruption and alleged sexual abuse/pardon/power of the president
124	Conservative politics and party loyalty versus progressive change
112	Billionaire and government taxation, spending, and wealth extraction concerns

Number of posts	LLM-generated label
106	Reproductive and Equal Rights (Women's Healthcare, Abortion Access, Voting Rights)
97	Anti-fascist and anti-nazi denunciation of white nationalism and alleged voter suppression
87	Tariffs driving higher prices and affordability concerns (Trump, economy, inflation)
84	Anti-Trump anger and criticism of political and corporate figures
74	Federal immigration enforcement and sanctuary cities (DHS/ICE/CBP, airports and international flights)
74	ICE and DHS detention enforcement, protests, and deportation demands
72	Debate over abortion (pro-life vs pro-choice) and definitions of fetal personhood and women's rights
68	Trans rights and women's safety amid transphobic political attacks
68	Abortion rights and pro-life vs pro-choice debate
68	Criticism of Donald Trump (idiocy, dementia, and related allegations)
63	LGBTQ+ Pride and Rights Debates (Marriage, Support, Trans Rights, Palestine)
62	Criticism of Trump supporters and alleged anti-intellectual, anti-minority voting behavior
56	US–Iran nuclear deal and regional war tensions involving Trump, Israel, and Netanyahu

We visualize the post as two-dimensional UMAP embeddings. We observe no clear linear separation between posts that were kept by human annotators versus removed. We also see no clear linear separation between posts that were unanimously labeled as keep or remove by human annotators versus posts labeled by split decision. This tells us that semantic meaning of posts by itself doesn't clearly determine keep vs. remove decisions. Topic and content alone is a weak separator. However, an exploratory 2-D representation itself doesn't tell us that there is no meaningful distinction between posts kept vs. removed by human annotators. This motivates our use of LLM-based feature extraction to mine for richer rhetorical and lexical patterns beyond what is available at a surface embedding level.

Method 2: LLM-based feature extraction

We also use LLMs to extract features. LLMs, unlike embeddings, can be steered to extract features from a richer set of categories, evaluating rhetorical, topical, lexical, and other patterns of the text.

We followed the following pipeline, inspired by current work out of Google

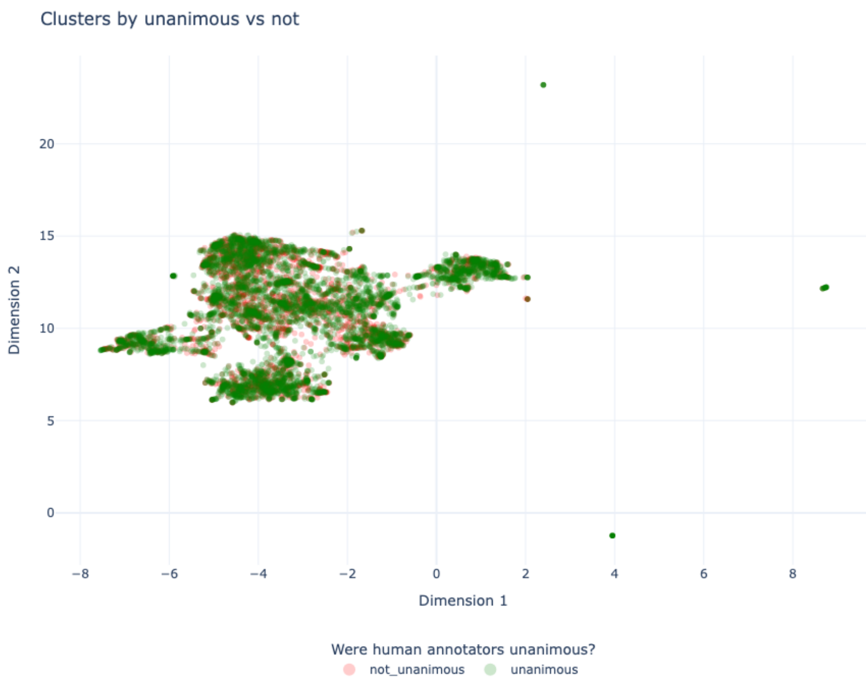

Figure 1: BERTopic UMAP projection by rater unanimity

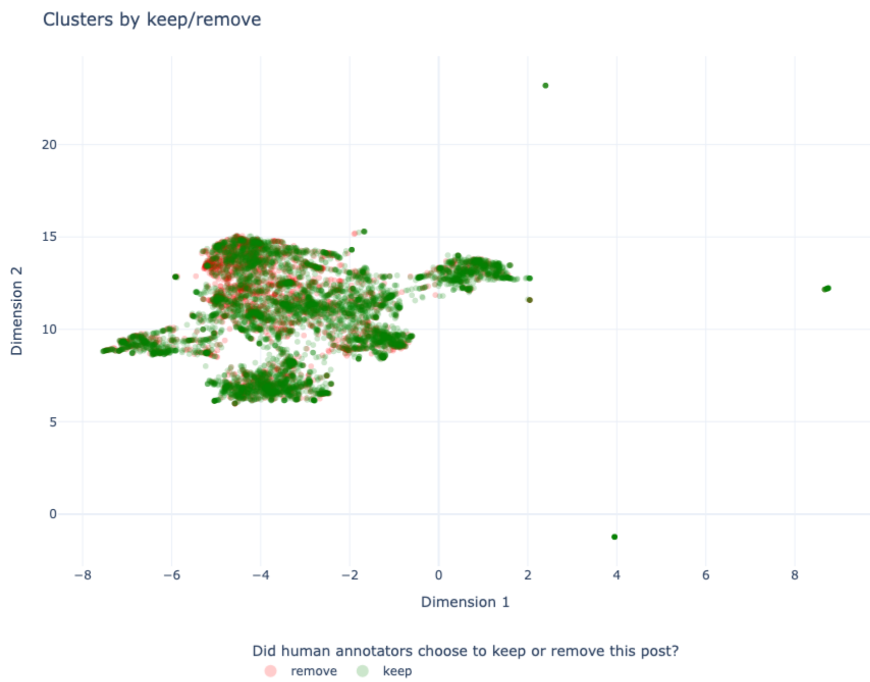

Figure 2: Clusters, colored whether human annotators kept or remove the posts

Figure 3: BERTopic UMAP projection by topic

DeepMind:

1. Give an LLM batches documents all at once and then ask it to generate features.
2. Get a semantic embedding of each generated feature.
3. Cluster the semantic embeddings.
4. Give an LLM a random sample of documents from each cluster and ask it to generate human-readable labels.

We use a systematic prompt to extract features across all these categories. We take a random sample of 500 posts removed by human annotators and 500 posts kept by human annotators. We then generated n=100 total LLM prompts for our initial data exploration. In each, we group n=10 posts (either all removed or all kept) and we ask the LLMs to generate features to explain why the human annotators could have chosen to keep or remove these posts.

The resulting clusters summarize recurring rhetorical and linguistic patterns in keep-rated and remove-rated posts. The topic analysis describes the policy areas and political themes in the corpus. The feature analysis describes rhetorical and linguistic patterns associated with each moderation decision. We use the two analyses as separate descriptive views of the data.

We observe a cleaner separation of features used to decide what posts to keep or what posts to remove using this new LLM-based method as compared to the BERTopic-based method. This is because our LLM prompting scheme can consider a richer set of criteria for what features to extract. We also notice a wider set of features uncovered for posts that were removed as compared to posts that were kept.

Comparing extracted features against reasons cited by participants

Both BERTopic and the LLM-based feature extraction generated a list of candidate features. We now compare them against features that were cited by study participants themselves. We asked our n=1,178 participants to answer, on a 1-7 Likert Scale, how much seeing the political mirror affected their moderation decision, as well as fill out a brief free-response section explaining their rationale. After filtering for noncompliance and invalid responses, we collected n=(TODO: insert amount) valid user responses.

We divided these into two groups, . . .

(Add the remaining analysis here).

Training a preliminary model with these features

. . .

(add latest results from latest run)

Figure 4: LLM feature cluster visualization (kept posts)

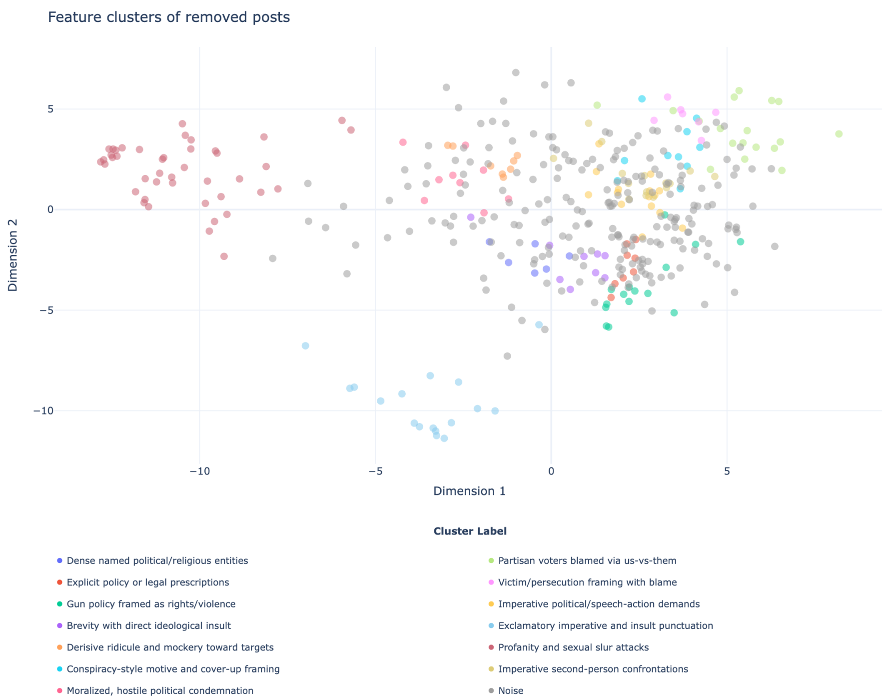

Figure 5: LLM feature cluster visualization (removed posts)

Arm	Accuracy	Precision	Recall	F1
control	0.6600	0.6932	0.5740	0.6280
prompt-tuned	0.6490	0.6114	0.8180	0.6997

(foop)